Exercise #1

We need to calibrate the link lengths of a planar 2R robot, whose nominal values are $\hat{l}_1 = \hat{l}_2 = 1$ [m]. All other kinematic parameters are assumed to be good enough. At four different Denavit-Hartenberg configurations $\boldsymbol{q}$, the following data (in [m]) for the position $\boldsymbol{p} \in \mathbb{R}^2$ of the robot end-effector are collected by an accurate external measurement system:

$$\begin{aligned} \boldsymbol{q}_a &= (0,0) & \Rightarrow \quad \boldsymbol{p}_a &= (2,0) \\ \boldsymbol{q}_b &= (\pi/2,0) & \Rightarrow \quad \boldsymbol{p}_b &= (0,2) \\ \boldsymbol{q}_c &= (\pi/4,-\pi/4) & \Rightarrow \quad \boldsymbol{p}_c &= (1.6925, 0.7425) \\ \boldsymbol{q}_d &= (0,\pi/4) & \Rightarrow \quad \boldsymbol{p}_d &= (1.7218, 0.6718). \end{aligned}$$

Provide the best estimate of the actual lengths l_1 and l_2 of the two robot links, using the above information. Is this calibration problem linear or nonlinear?

we need to compute the nominal DK:

$$p_a^{nom} = (2,0)^T \qquad p_b^{nom} = (0,2)^T \qquad p_c^{nom} = (1+\sqrt{2}/2, \sqrt{2}/2)^T \qquad p_d^{nom} = (1+\sqrt{2}/2, \sqrt{2}/2)^T$$

the matrix $\Delta\bar{\mathcal{E}} = (p_i - p_i^{nom})$ of the errors is:

$$\Delta\bar{\mathcal{E}} \begin{pmatrix} (0 & 0)^T \\ (0 & 0)^T \\ (-0.0146 & 0.0353)^T \\ (0.0146 & -0.0353)^T \end{pmatrix}$$

the regressor matrix is: $\Phi = \left(\frac{\partial x}{\partial \ell}\right) = \begin{pmatrix} c_1 & c_{12} \\ s_1 & s_{12} \end{pmatrix} \Rightarrow \bar{\Phi} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \\ \sqrt{2}/2 & 1 \\ \sqrt{2}/2 & 0 \\ 1 & \sqrt{2}/2 \\ 0 & \sqrt{2}/2 \end{bmatrix}$

Matlab

$$\Rightarrow \Delta\ell = \bar{\Phi}^{\#} \Delta\bar{\mathcal{E}} \Rightarrow \Delta\ell = (0.05, -0.05)^T \Rightarrow \begin{cases} \ell_1 = 1.05 \\ \ell_2 = 0.95 \end{cases}$$

error in ℓ

Exercise #2

A robot is driven by joint acceleration commands $\ddot{\boldsymbol{q}} \in \mathbb{R}^n$ which are kept constant for a (sufficiently small) sampling time T_c, i.e., $\ddot{\boldsymbol{q}}(t) = \ddot{\boldsymbol{q}}_k$, for $t \in [t_k, t_{k+1}) = [t_k, t_k + T_c)$. Thus, the next velocity at time $t = t_{k+1}$ can be expressed as $\dot{\boldsymbol{q}}_{k+1} = \dot{\boldsymbol{q}}(t_{k+1}) = \dot{\boldsymbol{q}}_k + T_c \ddot{\boldsymbol{q}}_k$. At time $t = t_k$, the robot is in the state $(\boldsymbol{q}_k, \dot{\boldsymbol{q}}_k)$ and has to realize a desired task acceleration $\ddot{\boldsymbol{r}}_{d,k} \in \mathbb{R}^m$, with $m < n$, being the task function $\boldsymbol{r} = \boldsymbol{f}(\boldsymbol{q})$. What is the expression of the command $\ddot{\boldsymbol{q}}_k$ that executes the task while minimizing the squared norm of the joint velocity at the *next* sampled instant t_{k+1}?

the acceleration is given by $\ddot{r} = \dot{J}\dot{q} + J\ddot{q}$, we denote $\dot{J}\dot{q} = n \Rightarrow \ddot{r} = n + J\ddot{q}$.

the joint velocity at the next time step is $\dot{q}_{k+1} = \dot{q}_k + T_c \ddot{q}_k$, the problem is the following:

$$\begin{cases} \min\limits_{\ddot{q}} & (\dot{q}_k + T_c\ddot{q})^T(\dot{q}_k + T_c\ddot{q}) = \dot{q}_k^T\dot{q}_k + T_c^2\ddot{q}^T\ddot{q} + 2T_c\dot{q}_k^T\ddot{q} \\ \text{s.t.} & J\ddot{q} = \ddot{r} - n \end{cases}$$

So the problem is to minimize this quadratic form, we can ignore $\dot{q}_k^T\dot{q}_k$.

$$\arg\min\ T_c^2\ddot{q}^T\ddot{q} + 2T_c\dot{q}_k^T\ddot{q} = \arg\min\ \ddot{q}^T\ddot{q} + \underbrace{2\frac{1}{T_c}\dot{q}_k^T}_{\text{let's denote } c}\ddot{q} = \arg\min\ \ddot{q}^T I \ddot{q} + c^T\ddot{q} =$$

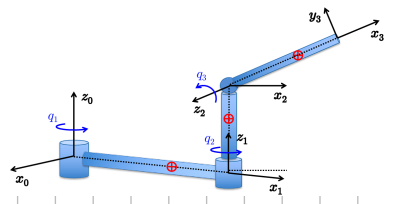
$$= \arg\min\left(\ddot{q} + \tfrac{1}{2}c\right)^T\left(\ddot{q} + \tfrac{1}{2}c\right)$$

$\Rightarrow$ "preferred direction" $= -\frac{1}{2}c = -\frac{1}{T_c}\dot{q}_k$ the solution is

$$\ddot{q}_k = -\frac{1}{T_c}\dot{q}_k + J^T(JJ^T)\left(\ddot{r} - \dot{J}\dot{q}_k - J\left(-\frac{1}{T_c}\dot{q}_k\right)\right).$$

Exercise #3

Consider the spatial 3R robot in Fig. 1. Using the D-H generalized coordinates defined therein, compute the robot inertia matrix $\boldsymbol{M}(\boldsymbol{q})$. Assume that the links have their center of mass on $\boldsymbol{x}_1$, $\boldsymbol{y}_2$, and $\boldsymbol{x}_3$, respectively, and that the barycentric link inertia matrices are diagonal, i.e., ${}^{i}\boldsymbol{I}_{ci} = \mathrm{diag}\,\{I_{ci,xx}, I_{ci,yy}, I_{ci,zz}\}$, $i = 1, 2, 3$.



	α	a	d	θ
1	0	l_1	0	q_1
2	$\pi/2$	0	l_2	q_2
3	0	l_3	0	q_3

I use the DH matrices:

$$ {}^0A_1 = \begin{pmatrix} C_1 & -S_1 & 0 & l_1C_1 \\ S_1 & C_1 & 0 & l_1S_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^1A_2 = \begin{pmatrix} C_2 & 0 & S_2 & 0 \\ S_2 & 0 & -C_2 & 0 \\ 0 & 1 & 0 & l_2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^2A_3 = \begin{pmatrix} C_3 & -S_3 & 0 & l_3C_3 \\ S_3 & C_3 & 0 & l_3S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} $$

I use the recurive method to compute T:

$$ {}^1\omega_1 = {}^0R_1^T \begin{pmatrix} 0 & 0 & \dot q_1 \end{pmatrix}^T = \begin{pmatrix} 0 & 0 & \dot q_1 \end{pmatrix}^T \qquad {}^1v_1 = {}^0R_1^T\left(\begin{pmatrix} 0 & 0 & \dot q_1 \end{pmatrix}^T \times \begin{pmatrix} l_1C_1 & l_1S_1 & 0 \end{pmatrix}^T \right) = \begin{pmatrix} 0 & l_1\dot q_1 & 0 \end{pmatrix}^T $$

due to the assumption: ${}^1r_{c1} = \begin{pmatrix} r_1 & 0 & 0 \end{pmatrix}^T$ so:

$$ {}^1v_{c1} = {}^1v_1 + {}^1\omega_1 \times {}^1r_{c1} = \begin{pmatrix} 0 & (l_1+r_1)\dot q_1 & 0 \end{pmatrix}^T $$

$$ {}^2\omega_2 = {}^1R_2^T\left({}^1\omega_1 + \begin{pmatrix} 0 & 0 & \dot q_2 \end{pmatrix}^T \right) = \begin{pmatrix} C_2 & S_2 & 0 \\ 0 & 0 & 1 \\ S_2 & -C_2 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot q_1 + \dot q_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \dot q_1 + \dot q_2 \\ 0 \end{pmatrix} \qquad {}^2r_{c2} = \begin{pmatrix} 0 & r_2 & 0 \end{pmatrix}^T \Rightarrow {}^2\omega_2 \times {}^2r_{c2} = 0 $$

$$ {}^2v_2 = {}^1R_2^T\left({}^1v_1 + \begin{pmatrix} 0 & 0 & \dot q_1 + \dot q_2 \end{pmatrix}^T \times \begin{pmatrix} 0 & 0 & l_2 \end{pmatrix}^T \right) = \begin{pmatrix} C_2 & S_2 & 0 \\ 0 & 0 & 1 \\ S_2 & -C_2 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ l_1\dot q_1 \\ 0 \end{pmatrix} = \begin{pmatrix} S_2 l_1 \dot q_1 \\ 0 \\ -C_2 l_1 \dot q_1 \end{pmatrix} $$

$$ {}^2v_{c2} = {}^2v_2 + {}^2\omega_2 \times {}^2r_{c2} = {}^2v_2 = \begin{pmatrix} S_2 l_2 \dot q_1 & 0 & -C_2 l_2 \dot q_1 \end{pmatrix}^T $$

$$ {}^3\omega_3 = {}^2R_3^T \begin{pmatrix} 0 & \dot q_1 + \dot q_2 & \dot q_3 \end{pmatrix}^T = \begin{pmatrix} S_3(\dot q_1 + \dot q_2) & C_3(\dot q_1 + \dot q_2) & \dot q_3 \end{pmatrix}^T $$

$$ {}^3v_3 = {}^2R_3^T\left({}^2v_2 + \begin{pmatrix} 0 & \dot q_1 + \dot q_2 & \dot q_3 \end{pmatrix}^T \times \begin{pmatrix} l_3C_3 & l_3S_3 & 0 \end{pmatrix} \right) = \begin{pmatrix} S_2C_3 l_1 \dot q_1 \\ l_3\dot q_3 - S_2S_3 l_1 \dot q_1 \\ -l_3C_3(\dot q_1 + \dot q_2) - C_2 l_1 \dot q_1 \end{pmatrix} $$

$$ {}^3v_{c3} = \begin{pmatrix} S_2C_3 l_1 \dot q_1 \\ l_3\dot q_3 - S_2S_3 l_1 \dot q_1 \\ -l_3C_3(\dot q_1 + \dot q_2) - C_2 l_1 \dot q_1 \end{pmatrix} + \begin{pmatrix} 0 & -\dot q_3 & C_3(\dot q_1 + \dot q_2) \\ \dot q_3 & 0 & -S_3(\dot q_1 + \dot q_2) \\ -C_3(\dot q_1 + \dot q_2) & S_3(\dot q_1 + \dot q_2) & 0 \end{pmatrix} \begin{pmatrix} r_3 \\ 0 \\ 0 \end{pmatrix} = $$

$$ \begin{pmatrix} S_2C_3 l_1 \dot q_1 \\ (l_3 + r_3)\dot q_3 - S_2S_3 l_1 \dot q_1 \\ -C_3(l_3 + r_3)(\dot q_1 + \dot q_2) - C_2 l_1 \dot q_1 \end{pmatrix} $$

Now i can compute the kinetic energy:

$$T_1 = \frac{1}{2} I_{c1zz} \dot{q}_1^2 + \frac{1}{2} m_1 (l_1 + r_1)^2 \dot{q}_1^2$$

$$T_2 = \frac{1}{2} I_{c2yy} (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1\dot{q}_2) + \frac{1}{2} m_2 l_2^2 \dot{q}_1^2$$

$$T_3 = \frac{1}{2}\left[(I_{c3xx} S_3^2 + I_{c3yy} C_3^2)(\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1\dot{q}_2) + I_{c3zz}\dot{q}_3^2\right] +$$
$$\frac{1}{2} m_3 \Big[S_2^2 C_3^2 l_1^2 \dot{q}_1^2 + (l_3 + r_3)^2 \dot{q}_3^2 + S_2^2 S_3^2 l_1^2 \dot{q}_1^2 - 2(l_3 + r_3) l_1 S_2 S_3 \dot{q}_1 \dot{q}_3$$
$$+ C_3^2 (l_3 + r_3)^2 (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1\dot{q}_2) + C_2^2 l_1^2 \dot{q}_1^2 + 2 l_1 (l_3 + r_3) C_2 C_3 (\dot{q}_1^2 + \dot{q}_1\dot{q}_2)\Big]$$

$$= \frac{1}{2}\left[(I_{c3xx} S_3^2 + I_{c3yy} C_3^2)(\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1\dot{q}_2) + I_{c3zz}\dot{q}_3^2\right] +$$
$$\frac{1}{2} m_3 \Big[l_1^2 \dot{q}_1^2 + (l_3 + r_3)^2 \dot{q}_3^2 - 2(l_3 + r_3) l_1 S_2 S_3 \dot{q}_1 \dot{q}_3 + C_3^2 (l_3 + r_3)^2 (\dot{q}_1^2 + \dot{q}_2^2 + 2\dot{q}_1\dot{q}_2)$$
$$+ 2 l_1 (l_3 + r_3) C_2 C_3 (\dot{q}_1^2 + \dot{q}_1\dot{q}_2)\Big]$$

i list the components of M as M_{ij}:

$$M_{11} = I_{c1zz} + m_1 (l_1 + r_1)^2 + I_{c2yy} + m_2 l_2^2 + I_{c3xx} S_3^2 + I_{c3yy} C_3^2 + m_3 (l_1^2 + C_3^2 (l_3 + r_3)^2 + 2 l_1 (r_3 + l_3) C_2 C_3$$

$$M_{22} = I_{c2yy} + I_{c3xx} S_3^2 + C_3^2 \left((l_3 + r_3)^2 m_3 + I_{c3yy}\right)$$

$$M_{33} = I_{c3zz} + m_3 (l_3 + r_3)^2$$

$$M_{12} = I_{c2yy} + (I_{c3xx} S_3^2 + I_{c3yy} C_3^2) + m_3 + C_3^2 (l_3 + r_3)^2 + l_1 (l_3 + r_3) C_2 C_3$$

$$M_{13} = -m_3 (l_3 + r_3) l_1 S_2 S_3$$

$$M_{23} = 0$$

Exercise #4

A planar 3R robot with unitary link lengths is commanded by a joint velocity $\dot{q} \in \mathbb{R}^3$ with components bounded as $|\dot{q}_i| \leq 2$ [rad/s], $i = 1, 2, 3$. The D-H joint variables have limited ranges specified by

$$q_1 \in [-\pi/2, \pi/2], \qquad q_2 \in [0, 2\pi/3], \qquad q_3 \in [-\pi/4, \pi/4].$$

At the configuration $\hat{q} = (2\pi/5, \pi/2, -\pi/4)$, the robot should move its end-effector horizontally with a speed $v_x = -3$ [m/s], while trying to keep the joints close to their midranges. Compute the value of the instantaneous joint velocity $\dot{q}$ that performs the Cartesian task while improving at best the criterion $H_{range}(q)$. Check if this joint velocity is feasible and, if not, perform the least end-effector task scaling to recover feasibility.

The function $H_{range}(q) := H$ is defined as follows:

$$H(q) = \frac{1}{6}\left[\frac{q_1^2}{\pi^2} + \left(3\frac{q_2 - \pi/3}{2\pi}\right)^2 + \left(2\frac{q_3}{\pi}\right)^2\right] \Rightarrow \nabla H = \frac{1}{6\pi}\left(\frac{2}{\pi} q_1, \frac{9}{2\pi} q_2 - \frac{3}{2}, \frac{8}{\pi} q_3\right)^T$$

The DK is

$$\begin{cases} C_1 + C_{12} + C_{123} \\ S_1 + S_{12} + S_{123} \end{cases} \Rightarrow J = \begin{pmatrix} -S_1 - S_{12} - S_{123} & -S_{12} - S_{123} & -S_{123} \\ C_1 + C_{12} + C_{123} & C_{12} + C_{123} & C_{123} \end{pmatrix}$$

$$\dot{q}_N = \nabla H(\hat{q}) = (0.042, 0.039, -0.106)^T$$

$$J(\hat{q}) = \begin{pmatrix} -2.151 & -1.2 & -0.891 \\ -1.096 & -1.405 & -0.454 \end{pmatrix}$$

$$\dot{\xi} = (-3, 0)^T$$

We decide to apply the Projected Gradient:

$$\dot{q}_{PG} = J^{\#}\dot{\xi} + (I - J^{\#}J)\dot{q}_N = (2.1512, -1.9261, 0.7677)^T$$

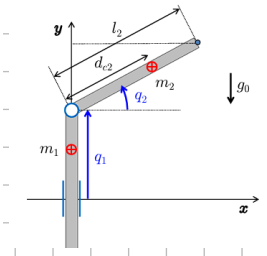
we scale the sol. to recover feasibility: $\frac{2}{2.1512}\dot{q}_{PG} = (2, -1.7907, 0.7137)^T$

Exercise #5

Figure 2 shows a PR robot and its inertia matrix, already expressed in terms of three dynamic coefficients a, b and c. The robot moves in a vertical plane. A task trajectory $y_d(t) \in \mathbb{R}$ is assigned to the coordinate y of the end-effector position. With the robot being at rest in the configuration $\bar{\boldsymbol{q}} = \begin{pmatrix} 1 & \pi/2 \end{pmatrix}^T$, provide the joint force/torque inputs $\boldsymbol{\tau}_A \in \mathbb{R}^2$ and $\boldsymbol{\tau}_B \in \mathbb{R}^2$ executing the desired task that instantaneously minimize, respectively,

$$H_A = \frac{1}{2}\|\boldsymbol{\tau}\|^2 \qquad \text{or} \qquad H_B = \frac{1}{2}\|\boldsymbol{\tau}\|^2_{M^{-2}(\bar{\boldsymbol{q}})}.$$

Which of the two solutions $\boldsymbol{\tau}_A$ and $\boldsymbol{\tau}_B$ has the largest first component in absolute value?



$$M(\boldsymbol{q}) = \begin{pmatrix} a & b\cos q_2 \\ b\cos q_2 & c \end{pmatrix} > 0$$

the DK is $\xi = q_1 + l_2 s_2 \Rightarrow J = \begin{pmatrix} 1 & l_2 c_2 \end{pmatrix} \Rightarrow J(\bar{q}) = \begin{pmatrix} 1 & 0 \end{pmatrix}$

the ξ componente of the CoM's are $p^{\xi}_{c1} = q_1 - d_{c1}$ $\quad p^{\xi}_{c2} = q_1 + d_{c2} s_2$ so

$U = g_0 \left(m_1 (q_1 - d_{c1}) + m_2 (q_1 + d_{c2} s_2) \right)$ with $g_0 \approx 9.8$

$\frac{\partial U}{\partial q_1} = g_0 (m_1 + m_2)$ $\quad \frac{\partial U}{\partial q_2} = g_0 m_2 d_{c2} c_2$ the model at rest in $\bar{q}$ is

$$\begin{cases} a\ddot{q}_1 + g_0(m_1 + m_2) = \tau_1 \\ c\ddot{q}_2 = \tau_2 \end{cases}$$

we have that $\ddot{\xi} = \dot{J}\dot{q} + J\ddot{q} \Rightarrow$

We solve $\ddot{\xi}_d = J\ddot{q}$ for $\ddot{q}$ while minimizing a given function, to obtain $\ddot{q}_d$, then we ploy it in the model and find (τ_1, τ_2).

$H_A)$ $\frac{1}{2}\|\tau\|^2 = \frac{1}{2}\left(a^2 \ddot{q}_1^2 + \underbrace{g_0^2 (m_1+m_2)^2}_{\alpha} + 2 g_0 (m_1+m_2) \ddot{q}_1 + c^2 \ddot{q}_2^2 \right) = \frac{1}{2} \ddot{q}^T \underbrace{\begin{pmatrix} a^2 + 2g_0(m_1+m_2) & 0 \\ 0 & c^2 \end{pmatrix}}_{W} \ddot{q} + \alpha$

$\Rightarrow$ we minimize $\frac{1}{2}\ddot{q}^T W \ddot{q}$ by using:

$$\ddot{q}_d = J_W^{\#} \ddot{\xi}_d = \begin{pmatrix} \ddot{\xi}_d & 0 \end{pmatrix}^T$$

$H_B)$ $\frac{1}{2}\|\tau\|^2_{M^{-2}} = \frac{1}{2}\tau^T M^{-2} \tau = \frac{1}{2}\begin{pmatrix} a\ddot{q}_1 + g_0(m_1+m_2) & c\ddot{q}_2 \end{pmatrix} M^{-2} \begin{pmatrix} a\ddot{q}_1 + g_0(m_1+m_2) \\ c\ddot{q}_2 \end{pmatrix} =$

$$M^{-2} = \begin{pmatrix} 1/a^2 & 0 \\ 0 & 1/c^2 \end{pmatrix} \Rightarrow H_B = \frac{1}{2}\left(\frac{1}{a^2}\left(a\ddot{q}_1 + g_0(m_1+m_2) \right)^2 + \ddot{q}_2^2 \right) =$$

$H_b = \ddot{q}_1^2 + \frac{1}{a^2} g_0^2 (m_1+m_2)^2 + 2\frac{1}{a} g_0 (m_1+m_2) \ddot{q}_1 + \ddot{q}_2^2$ we remove the constant

$$\Rightarrow \frac{1}{2}\left(\ddot{q}_1^2 + \underbrace{\frac{2}{a} g_0 (m_1+m_2)}_{\alpha} \ddot{q}_1 + \ddot{q}_2^2 \right) = \frac{1}{2}\ddot{q}^T I \ddot{q} + \begin{pmatrix} \alpha & 0 \end{pmatrix} \ddot{q} =$$

$$\frac{1}{2}\left(\ddot{q} - \frac{1}{2}\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)^T I \left(\ddot{q} - \frac{1}{2}\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$$

$\Rightarrow$ the sol is given by $\ddot{q} = \frac{1}{2}\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + J^T (J J^T)^{-1} \left(\ddot{\xi}_d - J \frac{1}{2}\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$

Exercise #6

For the same PR robot in Fig. 2, determine the gravity term $\boldsymbol{g}(\boldsymbol{q})$ in the dynamic model and define a tight upper bound $\alpha > 0$ on the norm of the square matrix $\partial \boldsymbol{g}(\boldsymbol{q}) / \partial \boldsymbol{q}$, for any value of $\boldsymbol{q}$.